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COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

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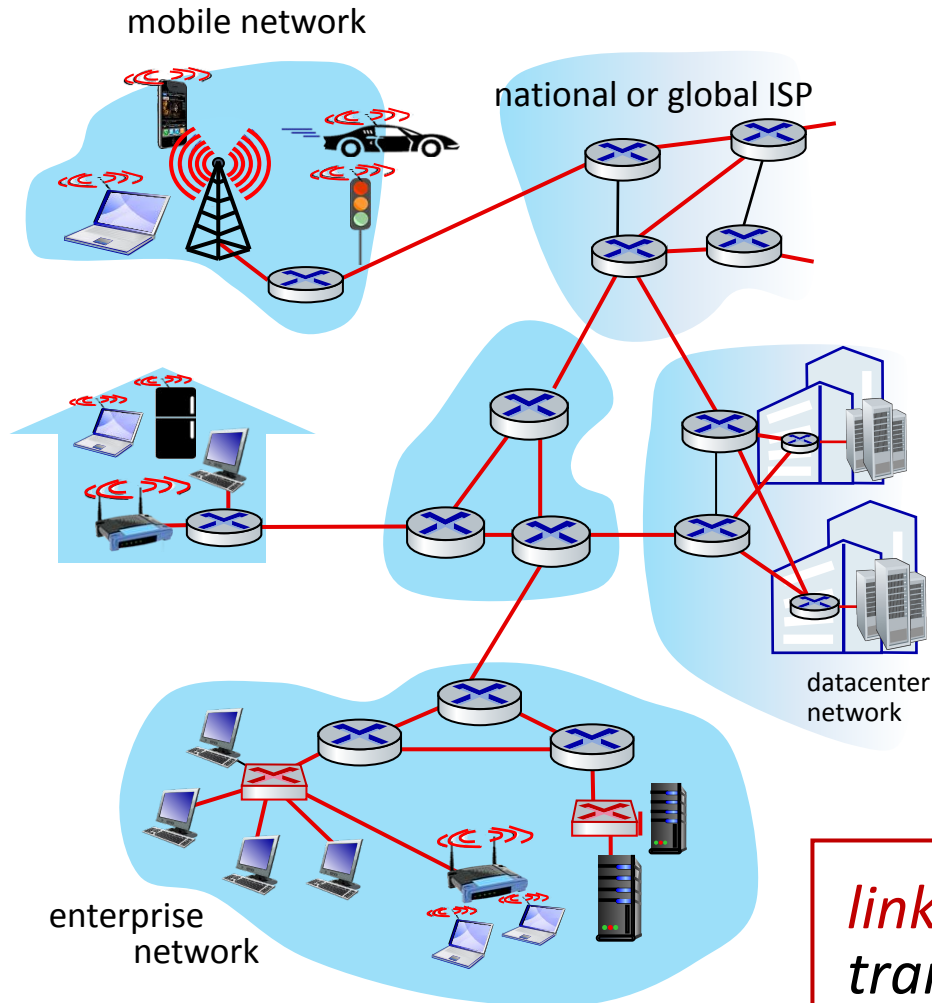
Link Layer and LAN

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- Introduction
- Error detection, correction
- Multiple access protocols
- LANs
 - addressing, ARP
 - Ethernet
 - switches
- A day in the life of a web request
- Physical layer
- Wireless LANs: IEEE 802.11

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Terminology:

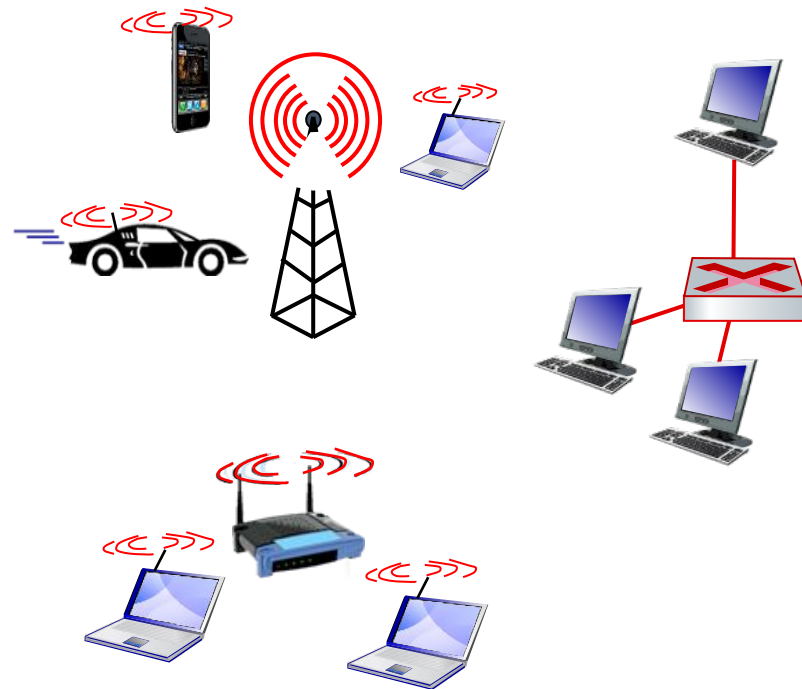
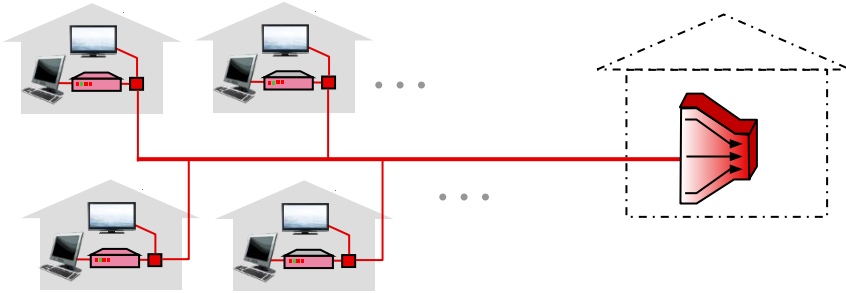
- Hosts, routers, switches, WiFi access points: Nodes
- communication channels that connect adjacent nodes along communication path: Links
 - wired
 - wireless
- layer-2 packet: *frame*, encapsulates datagram

link layer has responsibility of transferring datagram from one node to *physically adjacent* node over a link

- Datagram transferred by different link protocols over different links:
 - e.g., WiFi on first link, Ethernet on next link
- Each link protocol provides different services
 - e.g., may or may not provide reliable data transfer over link

transportation analogy:

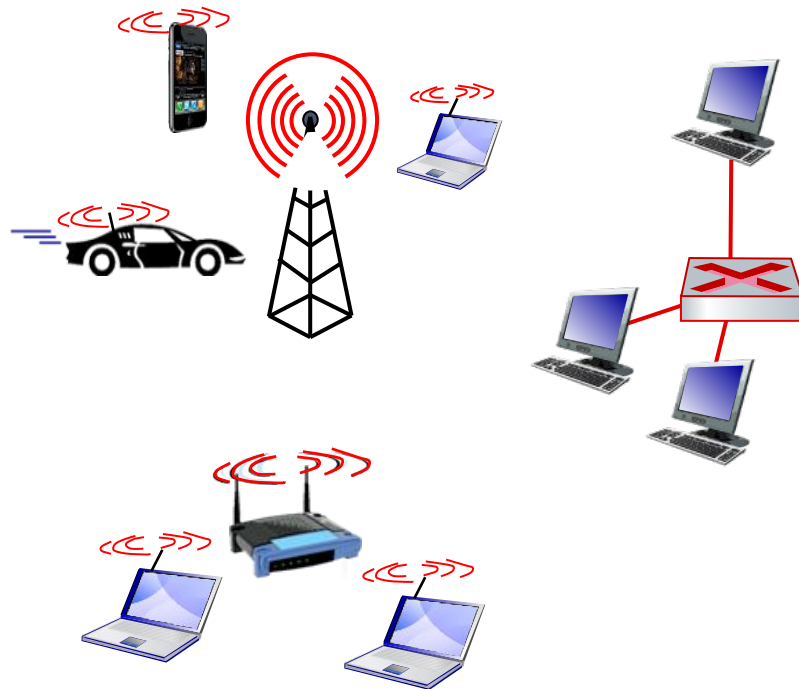
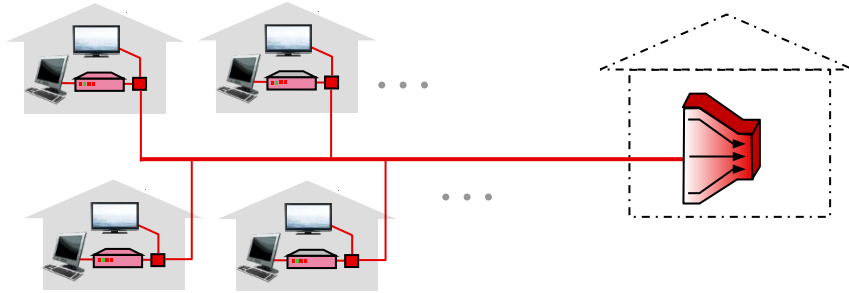
- trip from Princeton to Lausanne
 - limousine: Princeton to JFK airport
 - plane: JFK to Geneva
 - train: Geneva to Lausanne
- tourist = **datagram**
- transport segment = **communication link**
- transportation mode = **link-layer protocol**
- travel agent = **routing algorithm**



- **framing, link access:**
 - encapsulate datagram into frame, adding header, trailer
 - channel access if shared medium
 - “MAC” addresses in frame headers identify source, destination (different from IP address!)
- **reliable delivery between adjacent nodes**
 - we already know how to do this!
 - rarely used on low bit-error links (coax, fiber, etc)
 - wireless links: high error rates

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Link Layer: Services

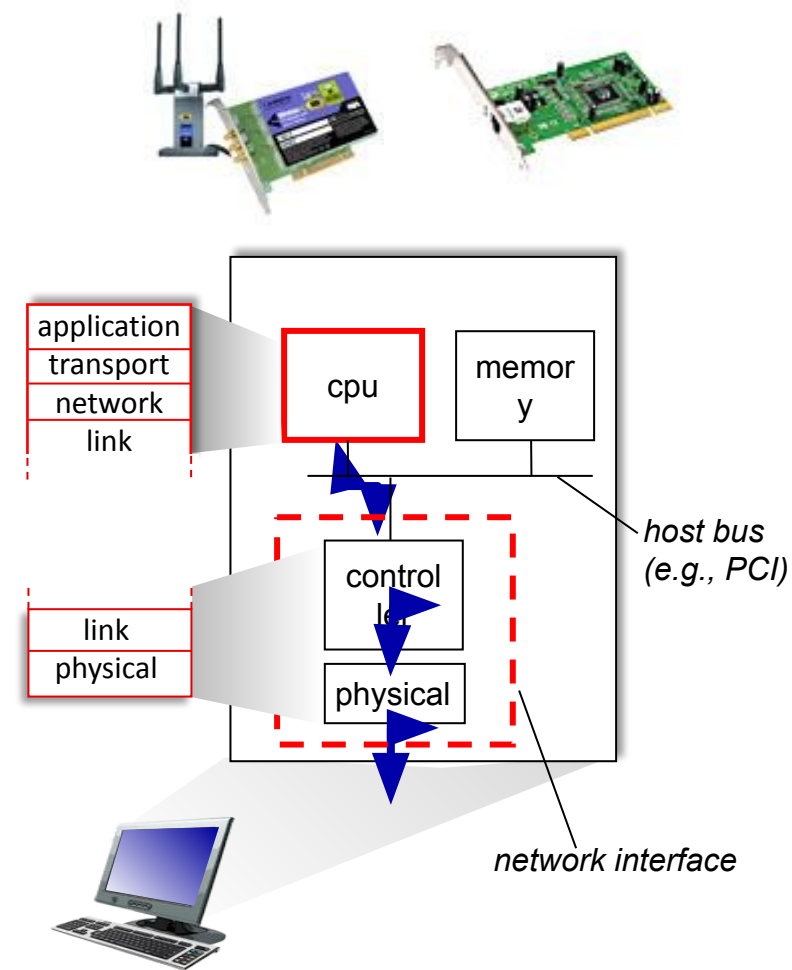


- **flow control:**
 - pacing between adjacent sending and receiving nodes
- **error detection:**
 - errors caused by signal attenuation, noise.
 - receiver detects errors, signals retransmission, or drops frame
- **error correction:**
 - receiver identifies *and corrects* bit error(s) without retransmission
- **half-duplex and full-duplex:**
 - with half duplex, nodes at both ends of link can transmit, but not at same time

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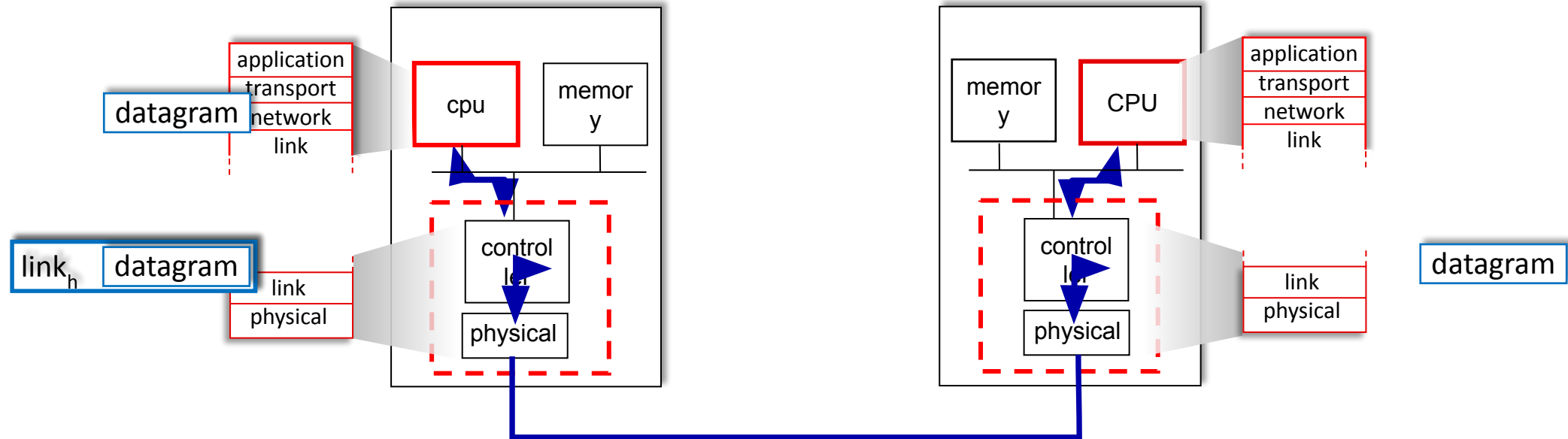
Where is the link layer implemented?

- In each-and-every host
- Link layer implemented in *network interface card* (NIC) or on a chip
 - Ethernet, WiFi card or chip
 - implements link, physical layer
- Attaches into host's system buses
- Combination of hardware, software, firmware



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Interfaces communicating



Sending side:

- encapsulates datagram in frame
- adds error checking bits, reliable data transfer, flow control, etc.

Receiving side:

- looks for errors, reliable data transfer, flow control, etc.
- extracts datagram, passes to upper layer at receiving side



THANK YOU

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